

Medical Image Analysis Tool (Segmentation + Classification)

Technical Analysis Document (max 10 pages)

Problem Statement

This project targets a real-world clinical workflow: automated screening support for chest X-rays.

The system performs two tasks jointly:

- Segmentation: delineate the lung fields (binary mask).
- Classification: predict a binary label (0/1) related to TB findings.

Why this problem is relevant

- Lung segmentation is a common pre-processing step that reduces background clutter and improves robustness.
- Binary screening helps triage large volumes of studies and prioritize suspicious cases.
- A combined segmentation+classification model allows feature sharing and provides a visual explanation (mask).

Scope and intended use

- This is a decision-support prototype, not a diagnostic device.
- Output should be reviewed by qualified clinical staff.
- Performance can change under domain shift (hospital/site/device/protocol).

Dataset: Montgomery County Chest X-ray (NLM) with lung masks; TB label from filename suffix (0=normal, 1=abnormal).

Evaluation split: test

Checkpoint: runs/bench_mps/checkpoints/best.pt

Methodology

End-to-end pipeline

1) Data acquisition and preprocessing

- Images are loaded as grayscale and denoised with a median filter (robust to salt-and-pepper artifacts).
- Each image and mask is resized to a fixed square resolution for batching.
- Intensity normalization is applied per image: $(x - \text{mean}) / \text{std}$ to reduce scanner/exposure variance.
- Data augmentation is applied only in training: random flips and 90-degree rotations to improve invariance.

2) Feature engineering / representation

- Learned representation with a CNN backbone (UNet encoder-decoder).
- Shared encoder features support both pixel-wise segmentation and image-level classification.

3) Core model logic (multi-task learning)

- Segmentation head outputs a single-channel logit map (lung vs background).
- Classification head uses global average pooling on bottleneck features + a linear layer to output 1 logit.

4) Optimization

- Loss = BCEWithLogits (classification) + BCEWithLogits (segmentation) + Soft Dice loss (segmentation).
- Optimizer: AdamW with weight decay to reduce overfitting.

5) Post-processing (segmentation)

- Probability thresholding converts logits to a binary mask.
- Morphological opening removes small spurious regions; closing fills small holes.
- Largest connected component selection enforces a single dominant lung region when appropriate.

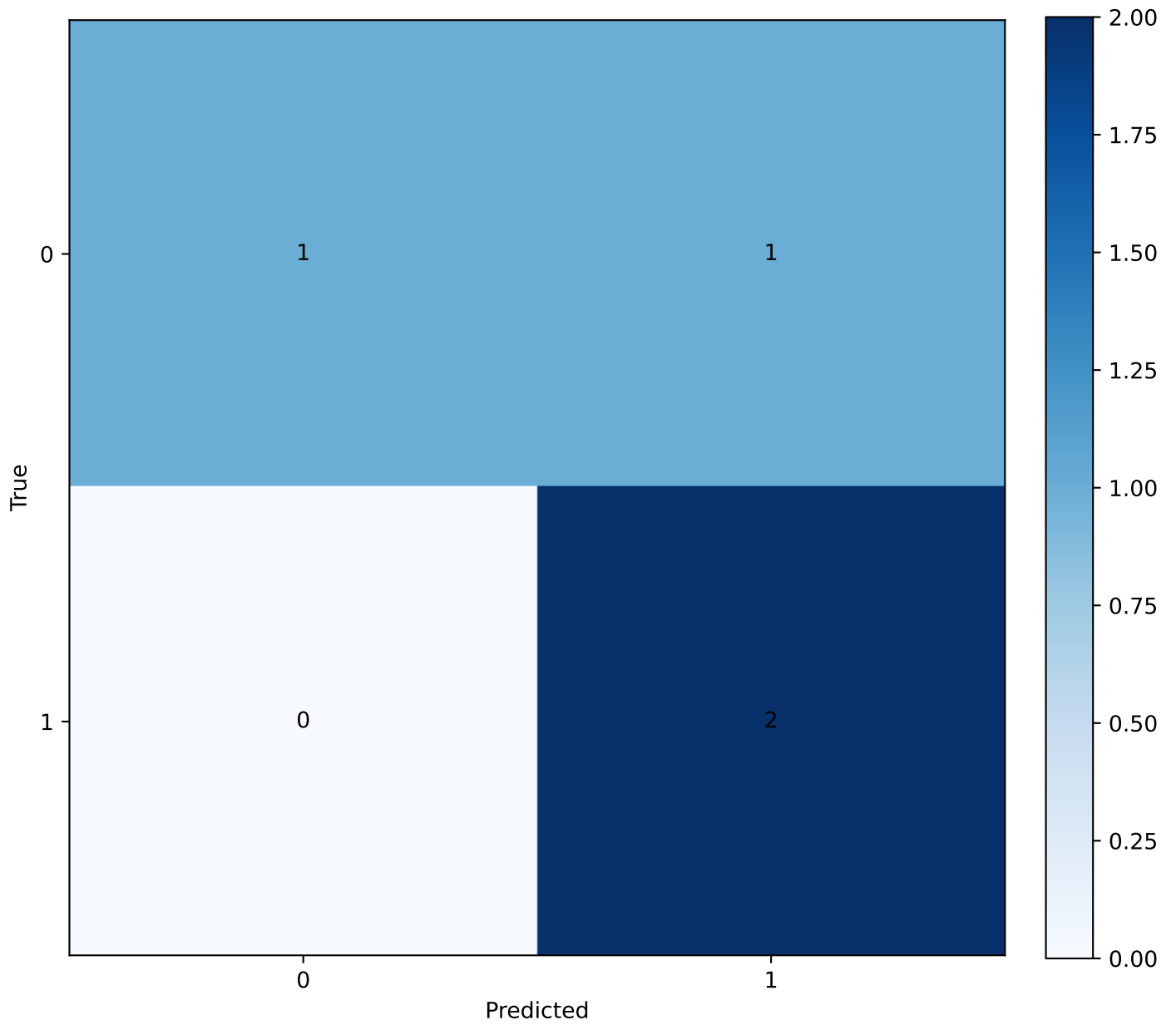
Implementation stack

- Python, OpenCV (pre/post-processing), PyTorch (model/training), scikit-learn (classification metrics)

Experimental Results (Table)

Metric	Value
cls_accuracy	0.7500
cls_f1	0.8000
cls_precision	0.6667
cls_recall	1.0000
loss_total	2.0513
seg_dice	0.4349
seg_miou	0.2808

Experimental Results (Confusion Matrix)



Failure Analysis

Observed and expected failure modes

Segmentation failures

- Poor contrast or strong acquisition artifacts may cause under-segmentation (missing peripheral lung).
- Strong field-of-view differences (cropped images) may shift the anatomy and degrade masks.
- Very bright/opaque regions can confuse boundaries and create holes or false regions.

Classification failures

- Class imbalance can lead to high recall but low specificity (more false positives).
- Domain shift (different hospital/site/device) can change texture statistics and reduce accuracy.
- Label noise: binary TB label is weak supervision and may not perfectly match visible findings.

Mitigations

- Expand augmentation (contrast/brightness) and use consistent intensity preprocessing.
- Tune the segmentation threshold and morphology kernels per dataset resolution.
- Calibrate classification threshold on validation data (optimize F1 or balanced accuracy).
- Fine-tune on a small labeled set from the target domain; validate cross-site when possible.

What to report during the oral exam

- Show a few qualitative examples: correct masks, typical failures, and how post-processing changes results.
- Explain why Dice/mIoU and F1/confusion matrix are appropriate for this task.

Ethical Considerations

Privacy and data handling

- Use de-identified data and avoid storing patient metadata in logs or filenames.
- Restrict access to raw data and follow institutional policies for medical data handling.

Bias, fairness, and generalization

- Performance may vary across demographic groups and acquisition devices/protocols.
- Report stratified metrics when subgroup information is available.
- Monitor false negative rate carefully due to the clinical cost of missed cases.

Clinical safety and limitations

- This tool is intended for decision support and requires human oversight.
- Communicate uncertainty: classification probabilities are not diagnoses.
- Document known failure cases and avoid over-claiming model capability.

Reproducibility

- Fix random seeds for comparable experiments and keep evaluation protocols consistent.
- Track configuration, checkpoints, and metrics files to enable auditability.